WALMART BUSINESS CASE

This analysis explores Walmart's Black Friday transaction data to understand customer purchasing behaviour and identify differences in spending across gender, marital status, and age groups. Exploratory Data Analysis (EDA), missing-value and outlier detection, and visualisations are used to understand the data. Confidence intervals and the Central Limit Theorem (CLT) are then applied to estimate average spending and support data-driven business recommendations.

Import libraries and load the data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats

sns.set_theme(style="whitegrid")

url = "https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/001/29

df = pd.read_csv(url)

# Standardise column names for easier coding
df.columns = df.columns.str.strip().str.replace(" ", "_")

print("Dataset loaded successfully")
print("Shape:", df.shape)
display(df.head())
```

```
Dataset loaded successfully
Shape: (550068, 10)
```

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Curre
0	1000001	P00069042	F	0-17	10	A	
1	1000001	P00248942	F	0-17	10	A	
2	1000001	P00087842	F	0-17	10	A	

The dataset contains transaction-level information about customers and their purchases during Black Friday.

Each row represents a transaction record, not necessarily a unique customer.

The Purchase column is the main variable for analysing customer spending.

1.A.Structure, data types and basic metrics

```
print("Rows and columns:", df.shape)
print("\nColumn names:")
print(df.columns.tolist())

print("\nData types and non-null counts:")
df.info()

print("\nStatistical summary:")
display(df.describe().T)

print("\nUnique values per column:")
display(df.nunique().to_frame("Unique_Count"))
```

```
Rows and columns: (550068, 10)

Column names:
['User_ID', 'Product_ID', 'Gender', 'Age', 'Occupation', 'City_Category', 'St

Data types and non-null counts:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   User_ID                     550068 non-null  int64
 1   Product_ID                  550068 non-null  object
 2   Gender                      550068 non-null  object
 3   Age                         550068 non-null  object
 4   Occupation                  550068 non-null  int64
 5   City_Category               550068 non-null  object
 6   Stay_In_Current_City_Years  550068 non-null  object
 7   Marital_Status              550068 non-null  int64
 8   Product_Category            550068 non-null  int64
 9   Purchase                    550068 non-null  int64
dtypes: int64(5), object(5)
memory usage: 42.0+ MB

Statistical summary:
```

	count	mean	std	min	25%	
User_ID	550068.0	1.003029e+06	1727.591586	1000001.0	1001516.0	10
Occupation	550068.0	8.076707e+00	6.522660	0.0	2.0	
Marital_Status	550068.0	4.096530e-01	0.491770	0.0	0.0	
Product_Category	550068.0	5.404270e+00	3.936211	1.0	1.0	
Purchase	550068.0	9.263969e+03	5023.065394	12.0	5823.0	

```
Unique values per column:
```

	Unique_Count
User_ID	5891
Product_ID	3631
Gender	2
Age	7
Occupation	21
City_Category	3
Stay_In_Current_City_Years	5
Marital_Status	2
Product_Category	20
Purchase	18105

Convert categorical attributes:

```
categorical_cols = [
    "Gender", "Age", "Occupation",
    "City_Category", "Stay_In_Current_City_Years",
    "Marital_Status", "Product_Category"
]

for col in categorical_cols:
    if col in df.columns:
        df[col] = df[col].astype("category")

print(df.dtypes)
```

```
User_ID                        int64
Product_ID                    object
Gender                      category
Age                         category
Occupation                  category
City_Category               category
Stay_In_Current_City_Years  category
Marital_Status              category
Product_Category            category
Purchase                       int64
dtype: object
```

The shape gives the total number of transaction records and attributes available for analysis.

Descriptive statistics summarise purchase amounts using the mean, median-related quartiles, standard deviation, minimum and maximum.

Demographic and product-category attributes are treated as categories because their values represent groups rather than continuous measurements.

Compare the mean and median of Purchase to get an initial idea of whether the spending distribution may be skewed.

1.B.Non-graphical analysis

```
for col in df.columns:
    print(f"\n--- {col} ---")
    print("Unique values:", df[col].nunique())

    if df[col].nunique() <= 20:
        print(df[col].value_counts(dropna=False))
```

```
--- City_Category ---
Unique values: 3
City_Category
B    231173
C    171175
A    147720
Name: count, dtype: int64

--- Stay_In_Current_City_Years ---
Unique values: 5
Stay_In_Current_City_Years
1     193821
2     101838
3      95285
4+     84726
0      74398
Name: count, dtype: int64

--- Marital_Status ---
Unique values: 2
Marital_Status
0    324731
1    225337
Name: count, dtype: int64

--- Product_Category ---
Unique values: 20
Product_Category
5     150933
1     140378
8     113925
11     24287
2      23864
6      20466
3      20213
4      11753
16      9828
15      6290
13      5549
10      5125
12      3947
7       3721
18      3125
20      2550
19      1603
14      1523
17       578
9        410
Name: count, dtype: int64

--- Purchase ---
Unique values: 18105
```

Check gender, marital status and age counts separately:

```
for col in ["Gender", "Marital_Status", "Age"]:
    if col in df.columns:
```

```
        print(f"\nCounts for {col}:")
        display(df[col].value_counts(dropna=False))
```

Counts for Gender:

Gender	count
M	414259
F	135809

dtype: int64

Counts for Marital_Status:

Marital_Status	count
0	324731
1	225337

dtype: int64

Counts for Age:

Age	count
26-35	219587
36-45	110013
18-25	99660
46-50	45701
51-55	38501
55+	21504
0-17	15102

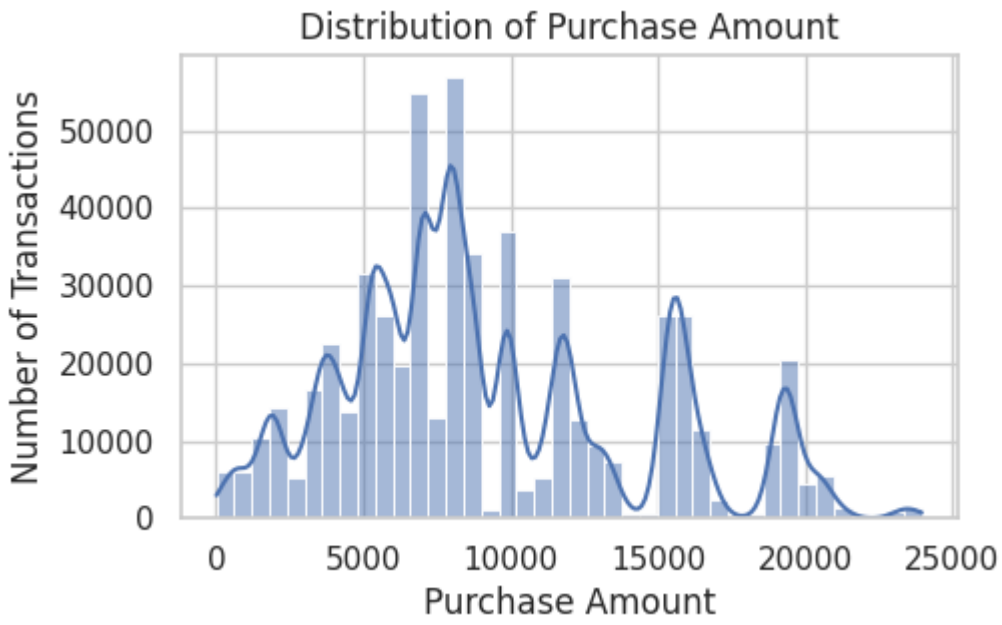
dtype: int64

1.C.Visual Analysis

```
plt.figure(figsize=(5, 3))
sns.histplot(data=df, x="Purchase", bins=40, kde=True)
plt.title("Distribution of Purchase Amount")
plt.xlabel("Purchase Amount")
plt.ylabel("Number of Transactions")
plt.show()

print("Mean purchase:", round(df["Purchase"].mean(), 2))
```

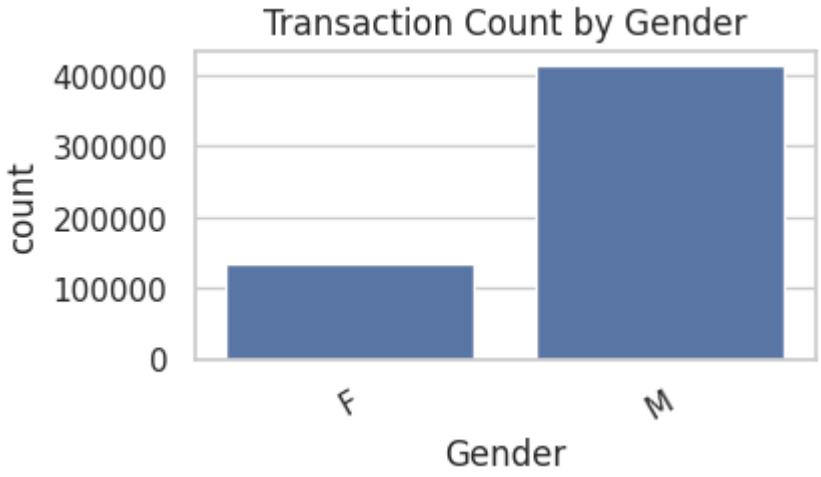
```
print("Median purchase:", round(df["Purchase"].median(), 2))
print("Skewness:", round(df["Purchase"].skew(), 2))
```

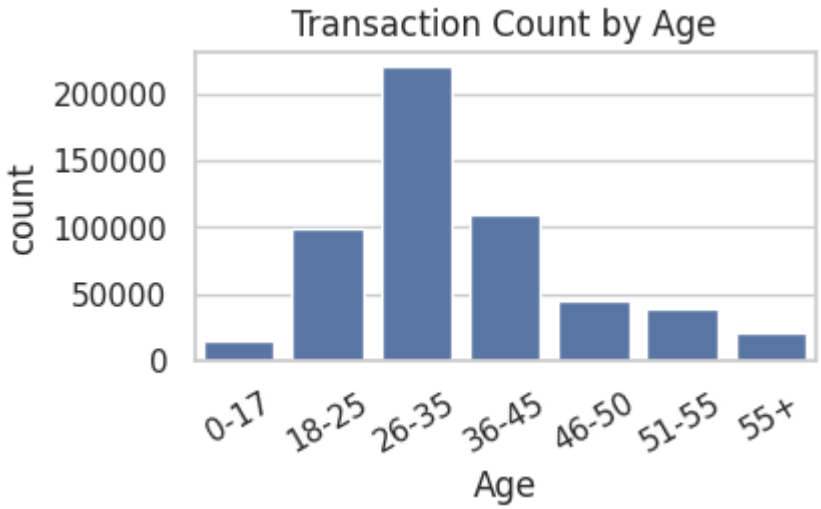


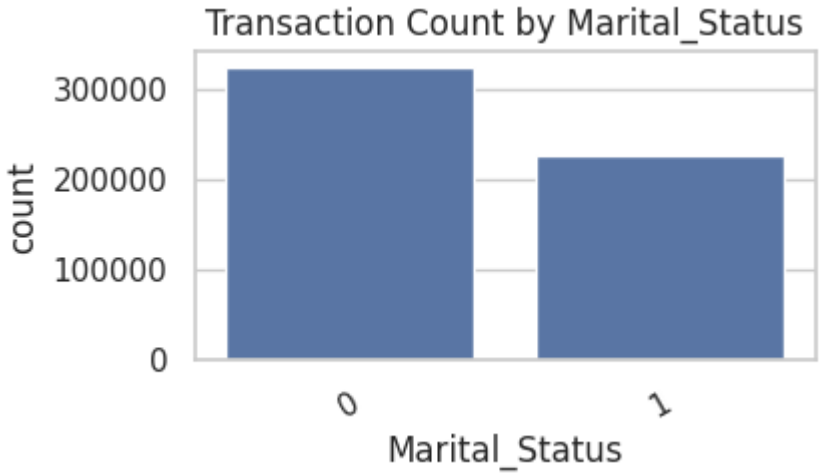

```
Mean purchase: 9263.97
Median purchase: 8047.0
Skewness: 0.6
```

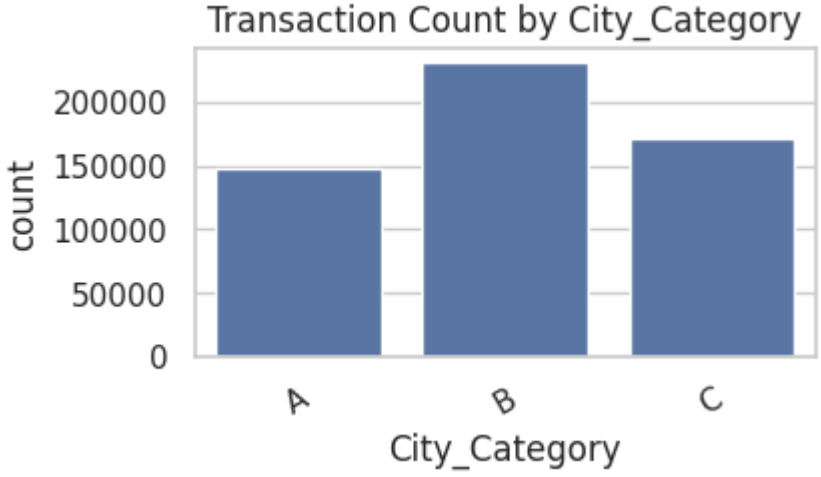
Count plots for categorical variables

```
for col in ["Gender", "Age", "Marital_Status", "City_Category"]:
    if col in df.columns:
        plt.figure(figsize=(4, 2))
        sns.countplot(data=df, x=col)
        plt.title(f"Transaction Count by {col}")
        plt.xticks(rotation=30)
        plt.show()
```


Transaction Count by Gender
count
400000
300000
200000
100000
0
F
M
Gender

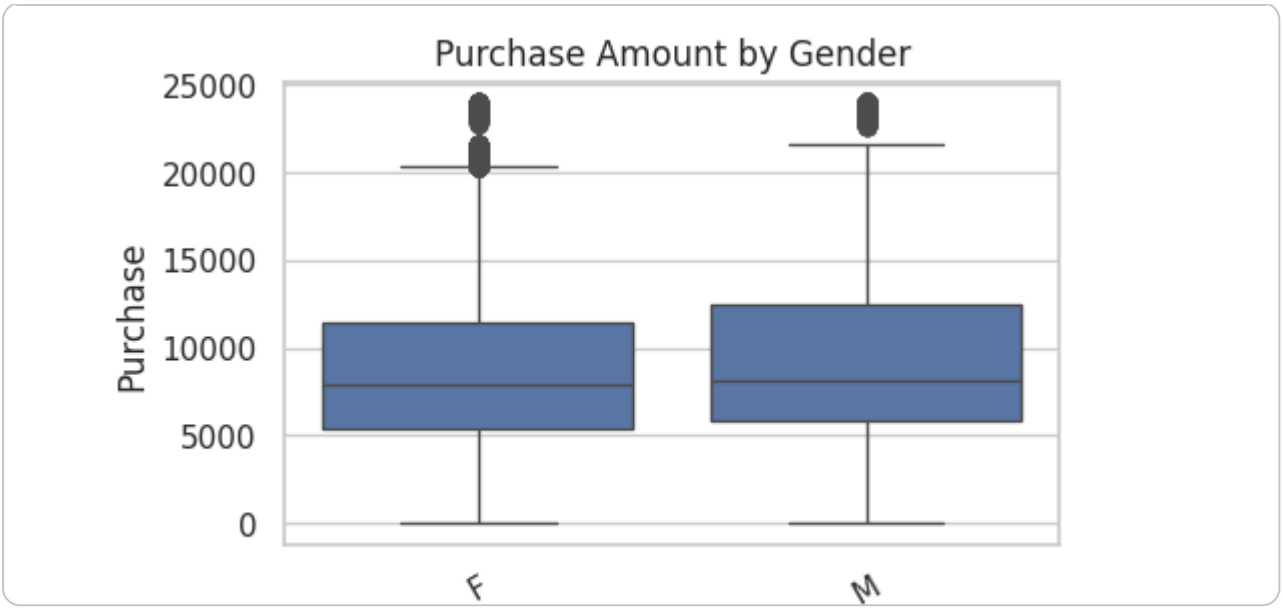

Transaction Count by Age
count
200000
150000
100000
50000
0
0-17
18-25
26-35
36-45
46-50
51-55
55+
Age


Transaction Count by Marital_Status
count
300000
200000
100000
0
0
1
Marital_Status


Transaction Count by City_Category
count
200000
150000
100000
50000
0
A
B
C
City_Category

Bivariate visual analysis

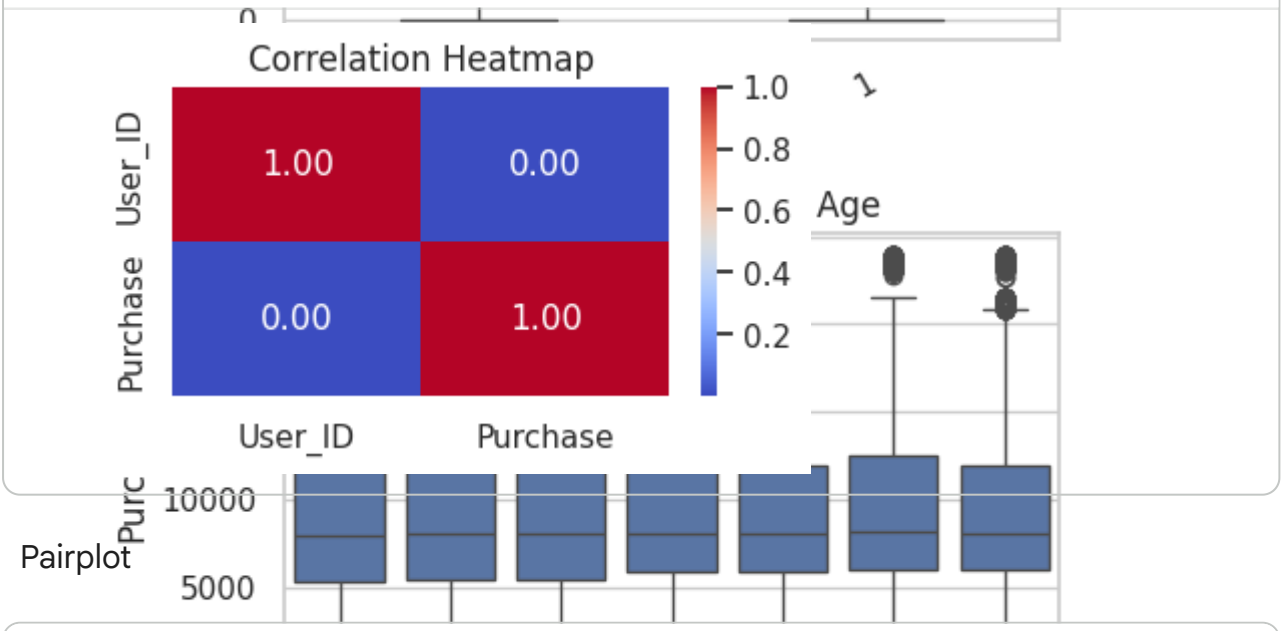
```
for col in ["Gender", "Marital_Status", "Age", "City_Category"]:
    if col in df.columns:
        plt.figure(figsize=(5, 3))
        sns.boxplot(data=df, x=col, y="Purchase")
        plt.title(f"Purchase Amount by {col}")
        plt.xticks(rotation=30)
        plt.show()
```



Correlation heatmap

```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(4, 2))
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Correlation Heatmap")
plt.show()
```

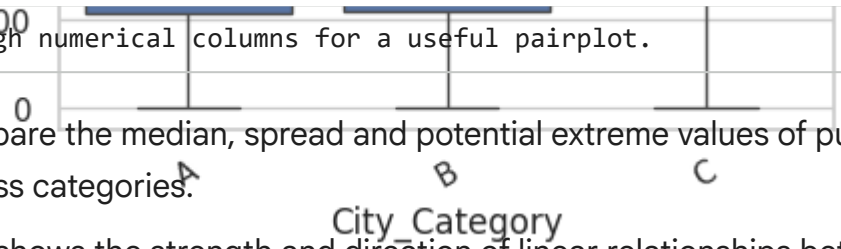


Pairplot

```
pair_cols = [
    col for col in
    ["Purchase", "Occupation", "Marital_Status", "Product_Category"]
    if col in df.columns
    and pd.api.types.is_numeric_dtype(df[col])
]
```

```
if len(pair_cols) >= 2:
    sample_size = min(2000, len(df))
    sns.pairplot(
        df[pair_cols].sample(sample_size, random_state=42)
    )
    plt.show()
else:
    print("Not enough numerical columns for a useful pairplot.")
```

```
Not enough numerical columns for a useful pairplot.
```



Boxplots compare the median, spread and potential extreme values of purchase amounts across categories.

The heatmap shows the strength and direction of linear relationships between numerical variables.

Pairplots help visualise pairwise patterns, but coded category values should not automatically be interpreted as continuous measurements.

A relationship between two variables does not necessarily mean one causes the other.

2.A.Missing-value detection

```
missing_count = df.isnull().sum()
missing_percent = (df.isnull().mean() * 100).round(2)

missing_summary = pd.DataFrame({
    "Missing_Count": missing_count,
    "Missing_Percentage": missing_percent
})

display(missing_summary)
print("Total missing cells:", df.isnull().sum().sum())
```

	Missing_Count	Missing_Percentage
User_ID	0	0.0
Product_ID	0	0.0
Gender	0	0.0
Age	0	0.0
Occupation	0	0.0
City_Category	0	0.0
Stay_In_Current_City_Years	0	0.0
Marital_Status	0	0.0
Product_Category	0	0.0
Purchase	0	0.0

```
Total missing cells: 0
```

Next steps: New interactive sheet

Missing-value analysis checks whether any attributes contain incomplete records.

If no missing values are found, no missing-value treatment is required.

If missing values exist, the treatment should depend on the column and the reason the values are missing.

Do not claim that the dataset has no missing values until the code confirms it.

2.B.Outliers Detection Use the boxplot and the IQR method for Purchase

```
Q1 = df["Purchase"].quantile(0.25)
Q3 = df["Purchase"].quantile(0.75)

IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[
    (df["Purchase"] < lower_bound) |
    (df["Purchase"] > upper_bound)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower bound:", lower_bound)
print("Upper bound:", upper_bound)
print("Potential outliers:", len(outliers))
print("Percentage of records:", round(len(outliers) / len(df) * 100, 2), "%"
```

```
plt.figure(figsize=(9, 4))
sns.boxplot(x=df["Purchase"])
plt.title("Outlier Detection for Purchase Amount")
plt.show()
```

```
Q1: 5823.0
Q3: 12054.0
IQR: 6231.0
Lower bound: -3523.5
Upper bound: 21400.5
Potential outliers: 2677
Percentage of records: 0.49 %
```



The IQR method identifies purchase amounts outside the calculated lower and upper bounds.

These records are potential outliers, not necessarily errors.

High purchase amounts may represent legitimate bulk or expensive purchases.

Outliers should be investigated before removing them; valid transactions should generally be retained for the business analysis.

3.Business Insights Based on Non-Graphical and Visual Analysis

3.A.Comments on the range of attributes

- The dataset contains customer demographic information, product categories, and purchase amounts from Walmart's Black Friday transactions.

- Numerical attributes such as Purchase, Occupation, and Product_Category can be examined using descriptive statistics, while demographic attributes such as Gender, Age, and City_Category help segment customers.
- The minimum, maximum, mean, and standard deviation of Purchase help us understand customers' spending range and variability.
- Unique-value counts show the variety of customer segments and product categories represented in the dataset.
- The dataset is transaction-level data, so a customer may appear in multiple rows because they can purchase multiple products.

3.B.Comments on variable distributions and relationships

- The Purchase distribution helps identify typical spending levels and whether a small number of transactions have unusually high purchase amounts.
- Frequency counts for Gender, Age, City_Category, and Marital_Status show how transactions are distributed across customer segments.
- Comparing purchase amounts by gender helps investigate whether average spending differs between male and female customers.
- Comparing purchase amounts by marital status and age group helps identify possible differences in spending across customer segments.
- Relationships between numerical variables can be explored using a correlation heatmap; correlation indicates association, not necessarily causation.

3.C.Comments for each univariate and bivariate plot write in summary The univariate analysis helps understand the distribution of purchase amounts and customer characteristics such as gender, age, marital status, city category, and product category. The bivariate analysis compares purchase behaviour across different customer segments using box plots and examines relationships between numerical variables using correlation heatmaps and pair plots. These visualizations help identify spending patterns, customer segments, and potential differences in purchase amounts. The insights provide a foundation for further statistical analysis and data-driven business recommendations for Walmart.

5. Insights Based on Exploration and CLT — Walmart Case Study

1. Distribution of variables and relationships

The exploratory data analysis (EDA) provides an overview of Walmart's Black Friday transactions, including customer demographics, product categories, and purchase

amounts. The distribution of purchase amounts helps identify common spending ranges, variability, and potential outliers. Customer attributes such as gender, age, marital status, and city category help segment transactions and compare spending patterns. These relationships provide a foundation for further statistical analysis using the Central Limit Theorem (CLT).

2. Comments on univariate and bivariate plots

- Purchase histogram: Shows the distribution and spread of transaction amounts, helping identify common spending ranges and potential skewness.
- Gender count plot: Compares the number of transactions from male and female customers.
- Age count plot: Identifies the age groups most represented in the dataset.
- Marital status count plot: Shows the transaction distribution between married and unmarried customers.
- City category count plot: Displays the distribution of transactions across different city categories.
- Purchase vs Gender box plot: Compares median spending, variability, and potential outliers across genders.
- Purchase vs Age box plot: Highlights differences in purchase distributions across age groups.

3. Generalizing the findings to the population using CLT

- Gender: Sample mean purchase amounts and confidence intervals can be used to estimate average spending for male and female customers.
- Marital status: Confidence intervals help estimate average spending for married and unmarried customers in the population.
- Age groups: Sample averages and confidence intervals help estimate average spending for customers in different age groups.
- Purchase amount: Repeated random sampling and calculating sample means illustrate how the sampling distribution becomes more stable as sample size increases.

4.Answering the Questions

Q1. Are women spending more money per transaction than men? Why or why not?

To answer this question, comparing the average purchase amount for female and male customers using the Gender and Purchase columns. Also comparing the median and standard deviation to understand typical spending and variability.

- Calculate the average purchase amount for each gender.

- Compare the averages to determine which group spends more per transaction in the sample.
- Use box plots to compare the median, spread, and outliers.
- Consider the number of transactions in each group when interpreting the results. Conclusion: The group with the higher average purchase amount spends more per transaction on average in the sample. However, this alone does not establish that the difference is statistically significant or explain why it occurs. Confidence intervals and the difference in means should be examined before making a final conclusion.

Q2.Confidence intervals and distribution of the mean of the expenses by female and male customers

A confidence interval estimates a plausible range for the population's average purchase amount based on the sample data. The Central Limit Theorem (CLT) explains why the sampling distribution of the sample mean tends to become approximately normal as the sample size increases, under appropriate conditions.

```
from scipy import stats
import numpy as np

for gender in df['Gender'].dropna().unique():
    values = df.loc[df['Gender'] == gender, 'Purchase'].dropna()
    mean = values.mean()
    se = stats.sem(values)
    lower, upper = stats.t.interval(
        0.95, df=len(values)-1, loc=mean, scale=se
    )

    print(f"{gender}:")
    print("Sample size:", len(values))
    print("Mean purchase:", round(mean, 2))
    print("95% CI:", (round(lower, 2), round(upper, 2)))
```

```
F:
Sample size: 135809
Mean purchase: 8734.57
95% CI: (np.float64(8709.21), np.float64(8759.92))
M:
Sample size: 414259
Mean purchase: 9437.53
95% CI: (np.float64(9422.02), np.float64(9453.03))
```

Observation: The confidence intervals provide an estimate of the average purchase amount for each gender. Larger samples generally produce narrower intervals when other factors remain comparable. The results help assess the precision of the estimated averages.

Q3.Are confidence intervals of average male and female spending overlapping? How can Walmart leverage this conclusion to make changes or improvements?

Compare the lower and upper bounds of the two confidence intervals obtained in Q2

- If the intervals overlap: The visual comparison alone does not establish whether the average spending differs significantly.
- If the intervals do not overlap: This provides evidence of a difference between the population means at the corresponding confidence level, under the usual assumptions.
- A more direct approach is to calculate a confidence interval for the difference between the two means and check whether it includes zero. Business recommendation: Walmart can use these findings to evaluate whether gender-specific promotions are justified. If a meaningful difference is supported by statistical analysis, Walmart can investigate product preferences, transaction patterns, and campaign response. Marketing decisions should also consider profitability and customer behaviour rather than gender alone.

Q4.Results for Married vs Unmarried customers Compare the purchase amounts of married and unmarried customers using the Marital_Status and Purchase columns. Calculate the mean and 95% confidence interval for each group.

```
for status, group in df.groupby('Marital_Status'):
    values = group['Purchase'].dropna()
```